ENGINEERING PORTFOLIO / 2026

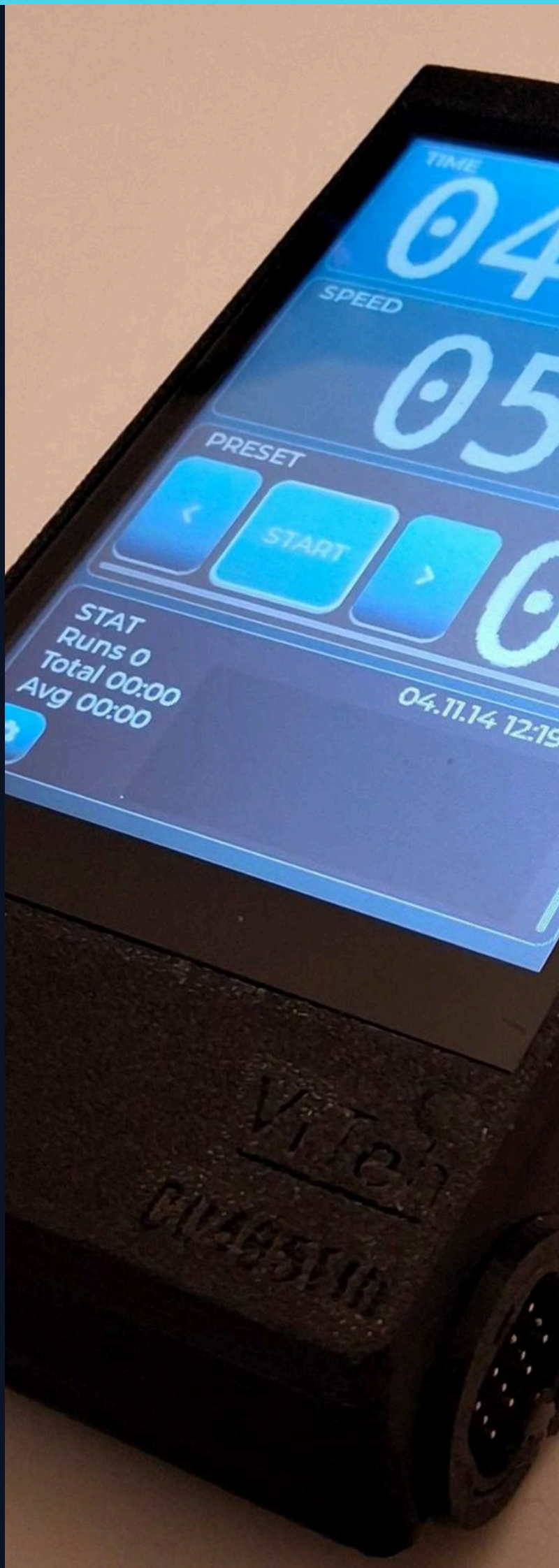
# Vitalii Krutikov

Embedded Systems Engineer
Hardware & PCB Designer
Software Developer

From circuit design and firmware to production-oriented prototypes, desktop tools, websites, Android apps and connected systems.

STM32 · ESP32 · C/C++ · C# · LINUX · PCB · IoT · AI

SELECTED WORK · VISIBLE PART OF WORK SINCE 2014



01 / PROFILE

# Engineering, end to end

Product development across embedded electronics, desktop software, web, Android, automation, networking and prototyping.

**PROFESSIONAL PROFILE**

Engineer and developer working since 2014 (12+ years), spanning embedded systems, PCB design, automation and PC software. Work includes sensor products, industrial controllers, telemetry devices, motor-control electronics and engineering utilities. Recent work adds websites and Android applications for private AI systems.

Experience: IT engineer at Adventx and Tesla-service, plus freelance projects.

## CORE CAPABILITIES

| | |
|---|---|
| 01 EMBEDDED | STM32 (F0–F7, G0/G3/G4), ESP32, AVR; C/C++; peripherals, drivers and integration. |
| 02 ELECTRONICS | Schematics, multilayer PCBs, prototyping, design reviews and enclosure CAD. |
| 03 CONNECTIVITY | CAN, Modbus RTU, RS-485, UART, USB, SPI, I²C, Ethernet, TCP/UDP, BLE and GSM. |
| 04 SOFTWARE | Desktop apps: C# (WinForms, WPF), Qt/C++, GTK, Swift; Python, Bash, diagnostics and automation. |
| 05 WEB & MOBILE | Fast static websites, web interfaces and admin panels (HTML/CSS/JS, Cloudflare); Android apps in Kotlin and Jetpack Compose. |
| 06 DELIVERY | Device configuration, troubleshooting, testing tools, technical documentation and visual instructions. |

02 / FEATURED PROTOTYPE

# Touchscreen control unit

Recent physical prototype photographed in November 2026.

### CUSTOM DEVICE

A compact, enclosed control system with an integrated color touchscreen, large numerical readouts, preset selection, start control and run statistics.

The photographed interface displays time, speed, preset navigation and status. The enclosure appears custom-made; its internal circuitry and firmware specifications are not established by the available photograph.

**DESIGN FOCUS** Human-machine interface · Enclosure integration · Device-level controls




Touchscreen control unit, prototype photo

02A / 3D MODELS

# Controller in 3D

Two generations of the touchscreen control unit as web-ready 3D models (GLB), exported from the original Blender project; the renders below come from those GLB files.



First version: small display, buttons, black cover




Latest version: large touchscreen, no buttons, ViTeh mark, round connector

**MODELS FOR THE WEB**

Lightweight GLB models (about 150 KB each) with PBR materials, the screen texture taken from a photo of the real unit; they plug into the interactive 3D product gallery on viteh.org.

SKILLS · Blender / glTF / WebGL

**FROM CAD TO PRODUCT PAGE**

The path from the enclosure design and PCB project (KiCad, Blender) to an interactive model that visitors can rotate, without third-party viewers or tracking.

SKILLS · enclosure CAD / 3D pipeline / web

03 / INDUSTRIAL CONTROL

# Sensors & automation

Selected projects documented with original product and system images.




Greenhouse screen controller

### GREENHOUSE CONTROLLER

Controller for opening and closing greenhouse screens, with a local display and associated control electronics. The original project archive includes connection illustrations and a video guide.




DMM-T2 sensor module

### DMM-T2 SENSOR MODULE

Temperature and humidity data acquisition modules with Modbus RTU communication, including variants for thermocouple inputs and different mechanical layouts.



Multi-channel temperature display




Ethernet-connected temperature monitor

04 / ELECTRONICS & PCB

# Custom hardware platforms

Original board imagery from previous hardware development projects.



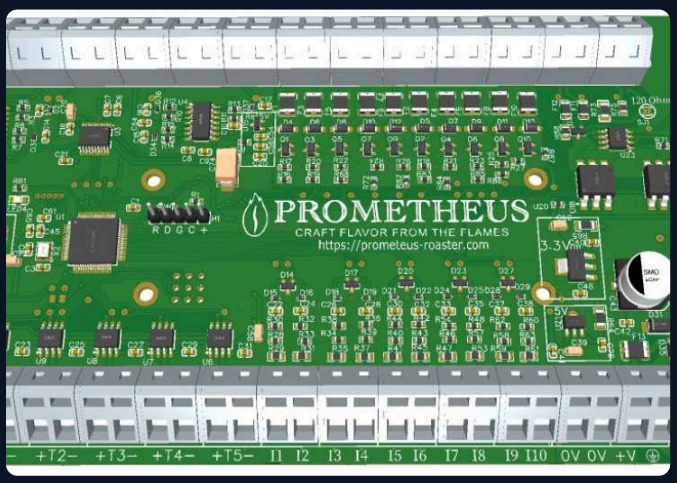

TryBrd / multi-layer stepper-control PCB

### MULTI-LAYER MOTION CONTROL

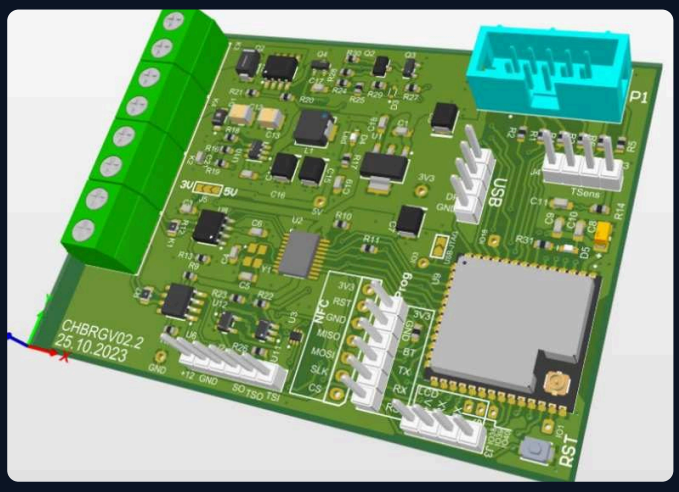
Designed a four-layer controller board for TMC5130A stepper motor drivers as part of a customer 3D-printing project; also developed basic motor-control firmware. Additional expansion-board work addressed multi-motor systems.

### OTHER BOARD WORK

I/O electronics, charging-station modules, GSM monitoring, and current ESP32-S3 gateway / Ethernet / RS-485 / CAN PCB work.




Custom I/O expansion board

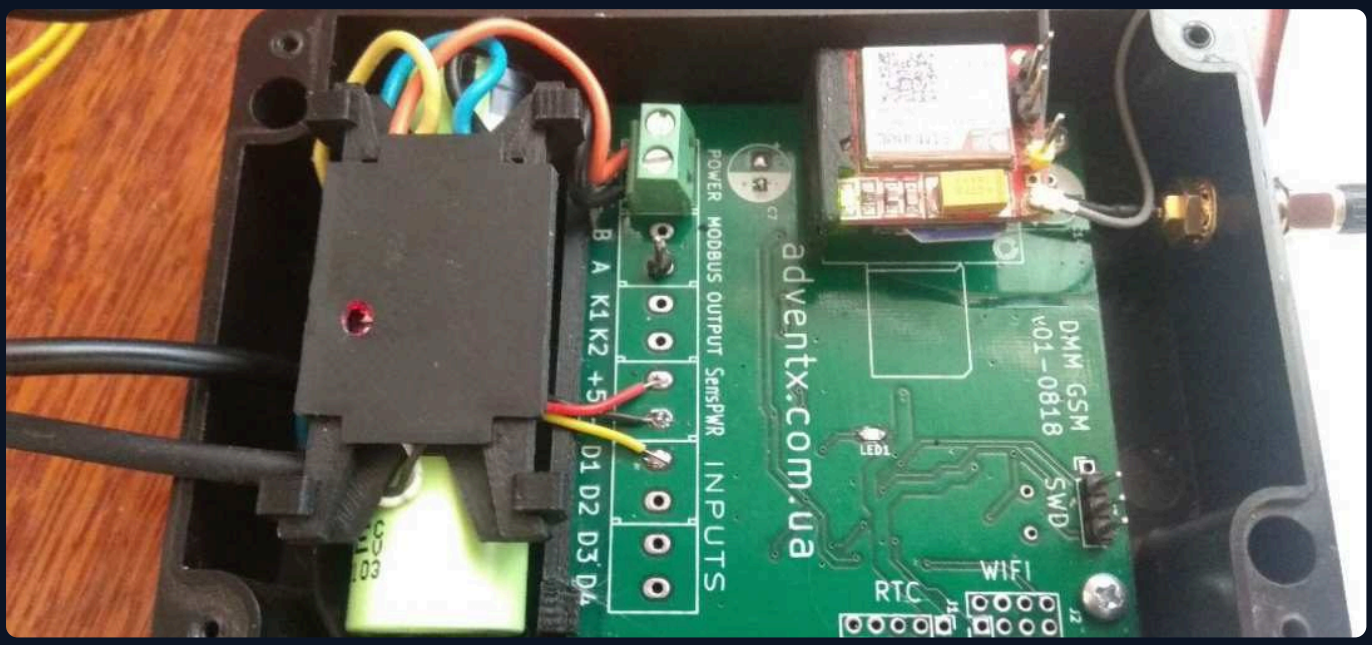



CHBRG charging-station device board

05 / CONNECTED DEVICES

# Telemetry & remote monitoring

Field electronics designed to collect information, report status and integrate with operator systems.




GSM event and sensor logger hardware

### GSM SENSOR LOGGER

Four sensor inputs, two relay outputs and backup power. The logger recorded sensor values and issued SMS alerts when configured thresholds were exceeded. It also uploaded data to a server for charting and remote configuration, with SMS-based commands.

### ADDITIONAL CONNECTED PRODUCTS

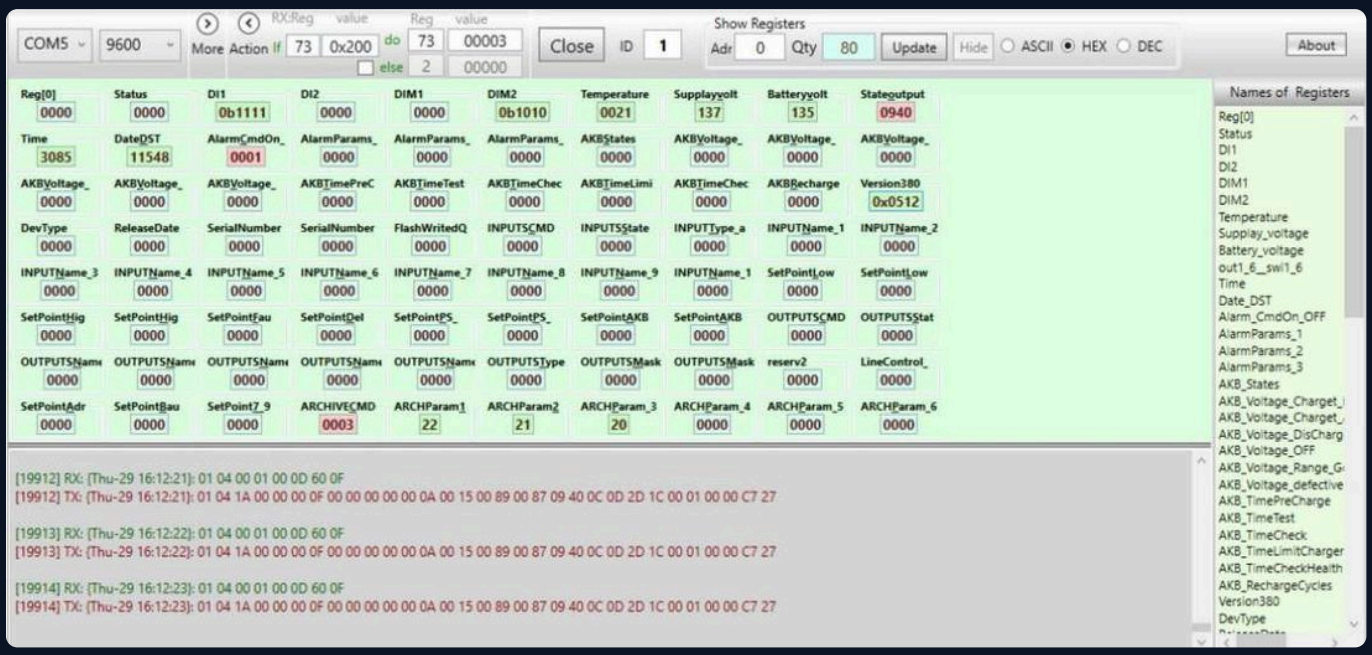
Broadcast watchdog: GSM-based radio-audio monitoring with loss-of-broadcast alerts and remote audio listening.

Event logger: PIR event timestamps collected to a PC utility, tabular viewer and CSV export; enclosure produced with 3D printing.

06 / SOFTWARE ENGINEERING

# Tools that support hardware

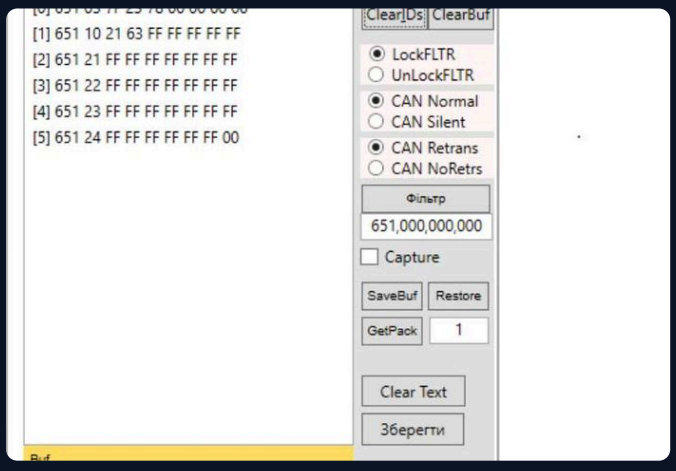
Desktop control applications, diagnostic utilities and cross-platform engineering workflows.



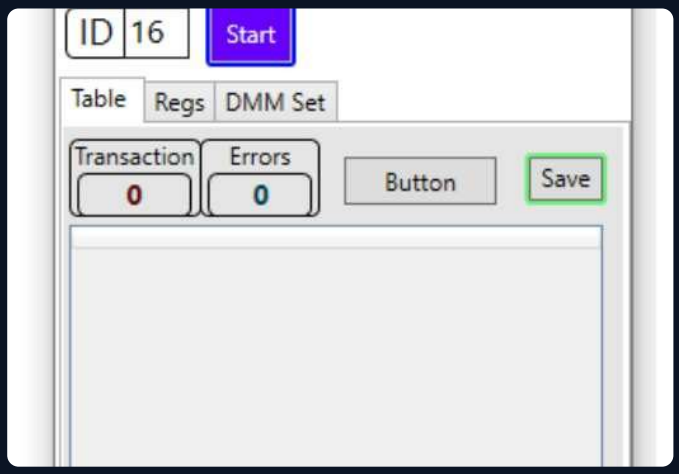

Modbus RTU diagnostic and register-monitoring utility

## DESKTOP TOOLS

C# software for sensor configuration and control, serial-port communication, Modbus analysis, CAN packet inspection and device diagnostics. Includes custom user interfaces for engineering and service tasks.




CAN analysis / packet utility

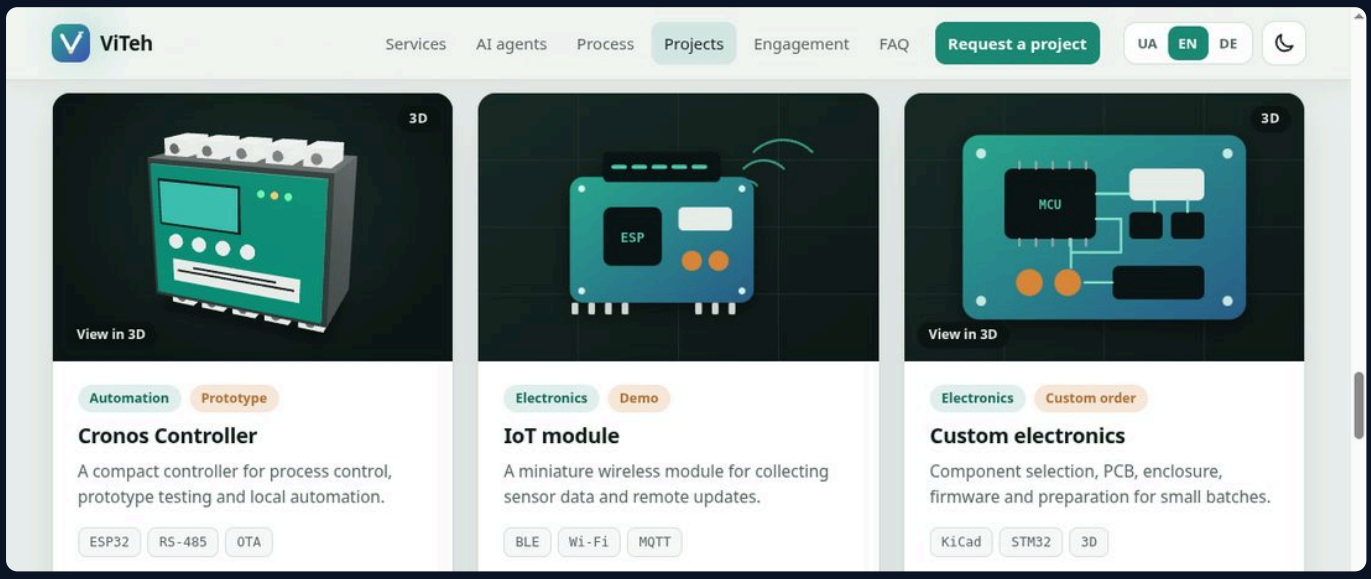



DMM configuration application

06A / WEB DEVELOPMENT

# Websites & web interfaces

Fast, private, multilingual sites, with the tools to run them.




viteh.org: product cards with a 3D view (UA / EN / DE, light and dark theme)

### STATIC SITES, NO TRACKERS

Static, no-build sites without third-party requests (no CDN, fonts or analytics), built to be GDPR-friendly, with legal pages for Germany and light/dark themes.

SKILLS · web design / performance / privacy

### CONTACT FORMS & INTEGRATIONS

Contact form with attachments served by a Cloudflare Worker and Email Routing, with spam protection and origin checks; hooks for mail and chat assistants.

SKILLS · Cloudflare Workers / forms / integrations

### 3D PRODUCT GALLERY

Interactive 3D models of hardware products (GLB exported from Blender), loaded lazily, one scene at a time, with a flat-image fallback.

SKILLS · 3D on the web / Blender / accessibility

### PRIVATE ADMIN & TEMPLATES

A private admin for the site: switch sections on or off, edit texts in three languages, build pages from templates, preview, then publish.

SKILLS · admin tooling / content workflow

06B / ANDROID APPLICATIONS

# Android client for private AI systems

Native Android application for the user's own servers and devices.

### CHAT & VOICE AGENT

Kotlin and Jetpack Compose app with a chat tab and a voice-agent tab: speech recognition, spoken replies, headset button controls and foreground services.

SKILLS · Kotlin / Compose / audio pipelines

### INTERPRETER & LECTURE MODE

Live translation and lecture capture with speech segmentation and voice detection, running against local speech and language models.

SKILLS · speech processing / streaming / UX

### FILES, MAIL & AGENT BOARD

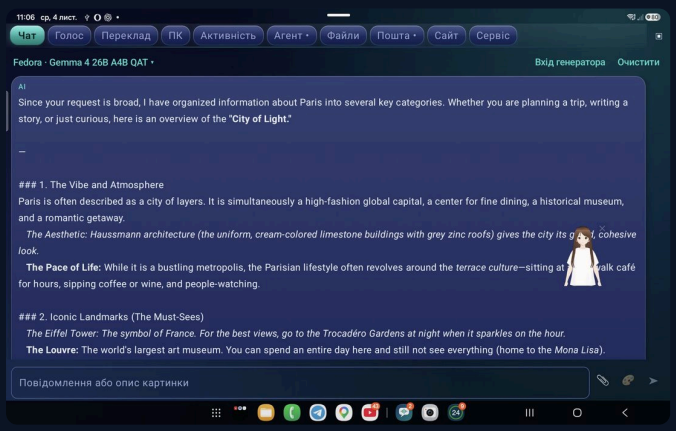
Files tab with previews, a mail tab with summaries and push-style notifications, and a board for agent tasks.

SKILLS · app architecture / notifications / protocols

### PRIVATE-NETWORK OPERATIONS

Routing over a private tailnet with fallbacks, automatic silent updates, on-device logs that can be uploaded for diagnostics, and a scripted release process.

SKILLS · networking / delivery / maintenance




Chat

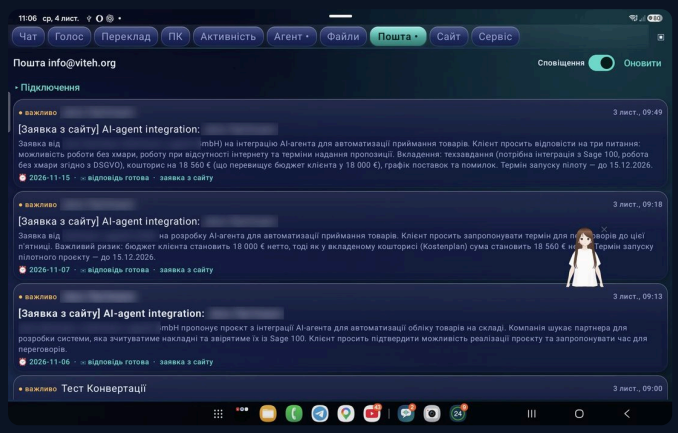



Live translation

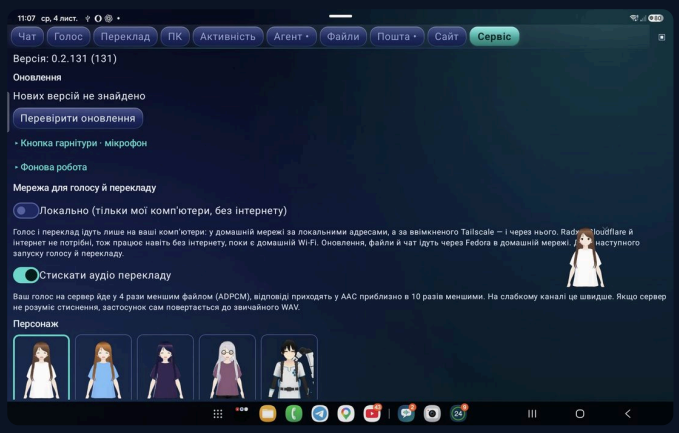
06C / AI CLIENT IN USE

# Private AI client: web and Android

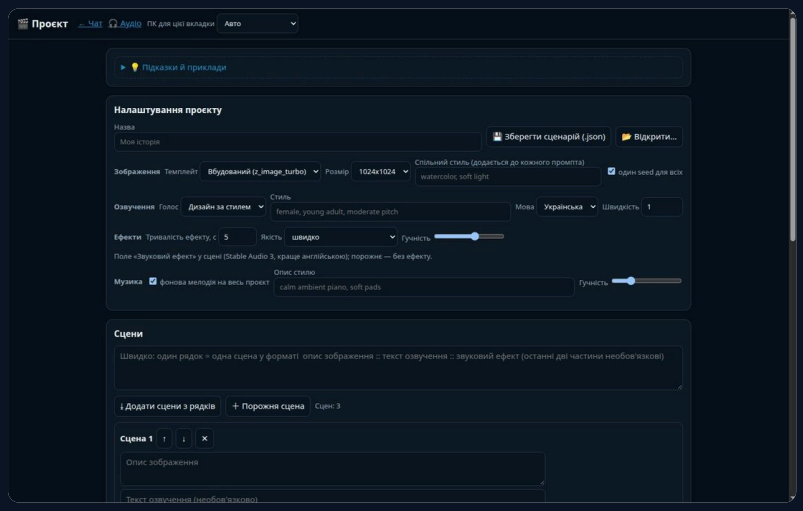
The same private AI system reached from a browser and from a native tablet app (screens with test data; private texts blurred).



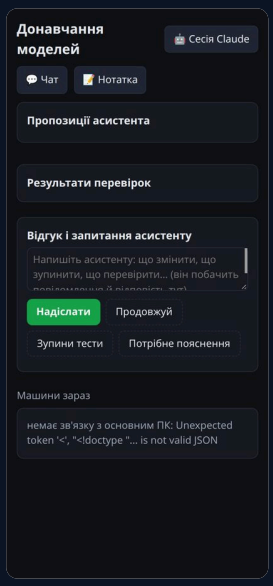

Mail agent (test messages)



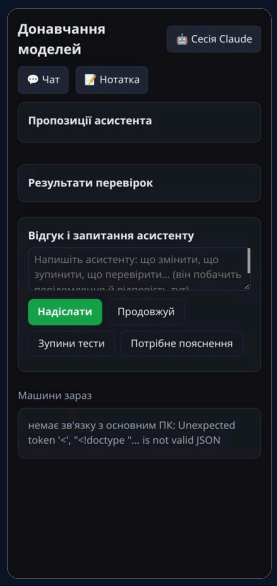

Service status




Web: project page, light theme




Web on a phone, dark theme




Web on a phone, light theme

07 / SERVICES

# What I offer

The same service names as on viteh.org, each backed by the work shown in this portfolio.

### Software development

Web apps, internal tools, APIs and desktop utilities tailored to your workflows.

Web and cloud · Integrations with your systems · Documentation and code handover

### Electronic devices

From schematic and PCB to firmware and enclosure — prototypes and small series.

Schematics and PCB layout · Firmware (ESP32, STM32, RP2040) · Enclosures and 3D printing

### AI-agent integration

Agents that read documents, talk to your systems and handle routine work under your control.

Connected to your data (RAG) · Tool and API calling · Cloud or local models

### IoT and automation

Sensors, gateways and dashboards: turning device data into a clear picture.

MQTT, Wi-Fi, BLE, LoRa · OTA firmware updates · Dashboards and alerts

### Prototypes and proof of concept

We test your idea quickly on a real device or demo before you invest more.

Working prototype within weeks · Clear success criteria · Scale-up recommendations

### Technical audit and consulting

An independent view on architecture, component choice, security and code quality.

Architecture review · Risk and cost assessment · Security by default

### Website development

Fast multilingual websites without trackers or third-party requests, forms with attachments, private admin tools and 3D product galleries.

Static sites without a build step · Forms, mail, Cloudflare Workers · 3D product models and a private admin

### Android apps

Native apps in Kotlin and Jetpack Compose for your own servers and devices: chat, voice, files, notifications.

Kotlin and Jetpack Compose · Voice, translation, background services · Automatic updates and private networks

### Custom apps built together with AI agents

Apps for almost any platform (Windows, Linux, macOS/iOS, Android, web), developed together with AI agents, with tests and support.

Prior experience: C# (WinForms, WPF), Qt, GTK, Swift · Current examples: Android client, Rust/Slint, web tools · Tests and support

### Local AI and voice

Running local language models on your own hardware, voice interfaces and fine-tuning.

Local LLMs, quantisation (GGUF) · Speech recognition and synthesis · Model fine-tuning

### Linux, networks and diagnostics

Linux/OpenWrt setup, remote access, and diagnosis of hangs, memory and network problems.

Linux, OpenWrt, systemd · Tailscale, Cloudflare Tunnel · Log analysis and debugging

### AI server: mail, website, documents

A private AI server for your tasks. Its parts are added as separate options: your own mail with SMTP and AI agents for mail, website and documents.

Option: own mail and SMTP · Option: AI agent for mail · Option: AI agents for website and documents

Desktop and cross-platform apps (C# WinForms/WPF, Qt, GTK, Swift): earlier projects on these platforms were not preserved and are not shown as case studies. They are offered as custom development together with AI agents; current examples are the Android client, a Rust/Slint application, web tools and a voice assistant.

07A / PROJECTS 2022–2025

# Projects 2022–2025

Boards, firmware and programs from the project archive. Details, roles and results on request.

Electronics · 2022–2023

### Prometheus: control board

A control board on the STM32G070 (KiCad) for coffee equipment, made to order: ModBus RTU, isolated RS-485, 7 analogue outputs 0–10 V, K-type thermocouple and 4–20 mA inputs, 10 digital inputs/outputs, DIN-rail mounting, 15–28 V supply.

STM32G070 · KiCad · ModBus RTU · RS-485

Electronics · 2022–2023

### Signal1: modules 380 and 382

Firmware for two STM32 boards (modules 380 and 382): menu, ModBus (archive, registers, alarms, calendar), a state diagram and test screenshots of a ModBus tool.

STM32 · ModBus · Firmware

EV charging · 2023–2025

### ChargeBridge: EV charging-station link

A board and software for talking to an electric-vehicle charging station over GB/T 27930-2015 (CAN) on ESP32/ESP32-S3. The board was designed in Altium; the device has no user interface, so 3D renders of the boards and the adapter are shown.

ESP32-S3 · CAN · GB/T 27930 · Altium

Electronics · 2024–2025

### Counter: V2.x boards

«Counter» boards, version V2.x, on an ESP32 with power switches; versions V2.3 and V2.4 are shown.

ESP32 · PCB · Altium

07A / PROJECTS 2022–2025

# Projects 2022–2025, continued

More boards, firmware and programs from the project archive.

Software · 2022

### CanHelper

A CAN/OBD tool for cars (Tesla and others) written in C# WPF for Windows, plus a separate ON/OFF mini helper.

C# · WPF · CAN · OBD

LVGL / HMI · 2025

### Toyota Prius hybrid-battery monitor

A graphical CAN-data monitor for the Prius on Qt6 for Linux (OrangePi Zero 2W): HV battery, module voltages, speed and temperature, simulation from a log. Also a touchscreen device on LVGL.

Qt6 · LVGL · CAN · Linux

LVGL / HMI · 2026-05

### LVGL smartwatch prototype

A smartwatch prototype on LVGL: a home screen with icons, a file list and a calendar.

LVGL · HMI

Software · 2015–2022

### Image and font tool for LCDs

A C# utility that converts images and fonts into formats for LCD displays.

C# · WinForms · LCD

07A / PROJECTS 2022–2025

# Projects 2022–2025, continued

More boards, firmware and programs from the project archive.

Software · 2025

### LANControl on Qt6

Control of a classroom or a network of PCs in Qt6/C++ for Windows: master and slave, locking, a Python editor.

Qt6 · C++ · Windows

Software · 2025–2026

### LANControl on Rust and Slint

The same classroom and PC-network control application, written in Rust with a Slint interface.

Rust · Slint · Windows

Software

### ModBus tools for DMM

C# utilities for ModBus and DMM modules, with charts.

C# · ModBus · DMM

Roles, customers and results are available on request.

08 / EMBEDDED R&D

# Embedded Systems & HMI

Research, experiments and prototypes are part of my applied engineering expertise, not just future product ideas.

### ESP32-S3 / LVGL GRAPHICAL INTERFACES

Implementation and evaluation of LVGL 9.x touchscreen user interfaces; display bring-up using CO5300, SH8601, NV3041A and ST7789 controllers; SPI/QSPI display integration, FT3168 touch and PSRAM constraints.

SKILLS · UI architecture / display drivers / HMI debugging

### POWER MANAGEMENT & WIRELESS CONTROL

ESP32-C6 + ESP-NOW servo-control experiments with wake, transmission and sleep cycles. Evaluated deep-sleep and light-sleep behavior, power consumption, PWM control and responsiveness.

SKILLS · low-power design / telemetry / measurement

### MCU PLATFORMS & BOARD BRING-UP

Experience exploring STM32 and ESP32 alongside RISC-V and Linux-capable boards including Milk-V Duo, LicheeRV Nano and RK3506-based hardware. Evaluated peripheral capabilities and development workflows.

SKILLS · platform evaluation / firmware integration

### EDGE HARDWARE & PCB ARCHITECTURE

Design investigation of connected microcontroller boards with CAN, RS-485, Ethernet, digital I/O and sensor interfaces; experience considering board layout, interoperability and physical integration.

SKILLS · schematics / interface selection / prototyping

09 / SYSTEMS R&D

# Linux, Networks & Diagnostics

Hands-on system integration, debugging, protocol analysis and real-device experimentation.

### LINUX / OPENWRT NETWORKING

Configured and investigated OpenWrt gateways, virtualized routers, IPv4 routing, LAN bridges, Tailscale subnet routing, Cloudflare Tunnel and remote management workflows.

SKILLS · networking / firewall logic / remote access

### SINGLE-BOARD COMPUTERS & SERVICES

Raspberry Pi and embedded Linux platform investigations: storage and boot planning, SSH, services, Docker-based workflows and device-oriented Linux automation.

SKILLS · Linux administration / Bash / systems integration

### AUTOMOTIVE CAN BUS & DIAGNOSTICS

Practical Toyota Prius hybrid-battery repair and diagnostic work with CANable V2, Python-CAN, ECU request/response traffic, battery-block voltage and regeneration/load measurements.

SKILLS · CAN / diagnostics / data interpretation / hardware repair

### DESKTOP & DEVICE INTEGRATION

Experience solving hardware/software integration problems across Linux and Windows, including device interfaces, remote desktop, printing services, USB, serial tools and firmware processes.

SKILLS · troubleshooting / integration / automation

10 / COMPUTE R&D

# Local AI & GPU Engineering

Applied investigations that developed demonstrable technical knowledge in model deployment and heterogeneous computing.

### LOCAL LLM DEPLOYMENT

Hands-on operation and testing of coding and general-purpose local models, including GGUF quantized deployments. Compared model quality, quantization, memory footprint and latency trade-offs.

SKILLS · inference deployment / benchmarking / evaluation

### AMD ROCm / MEMORY MANAGEMENT

Troubleshot ROCm model-loading behavior on an AMD unified-memory system, including mmap versus mlock loading, memory limits and model residency. Documented successful tool-call tests in personal experiments.

SKILLS · ROCm / runtime debugging / memory profiling

### MULTI-GPU / EXPERT PLACEMENT

Explored NVIDIA and AMD compute configurations, GPU VRAM limits, CPU/GPU offload, MoE active-expert placement, partitioning and the potential impact on inference throughput.

SKILLS · hardware-aware optimization / performance analysis

### AI-ASSISTED ENGINEERING WORKFLOWS

Experimented with local coding assistants, tool-calling reliability, scripting, Git-based workflows and repeatable technical checks for embedded and software development.

SKILLS · developer tooling / validation / agent workflows

11 / COLLABORATION

# Building real systems.

I work across the boundary of hardware and software: from embedded interfaces, sensing and communication to PC utilities, websites, Android applications, troubleshooting tools and physical product prototypes.

## RELEVANT ROLES

- Embedded Systems Engineer / Firmware Developer
- Electronics & PCB Designer / IoT & Automation Developer
- Hardware–Software Integration Engineer
- Web & Android Application Developer

## TECHNOLOGY

C · C++ · C# (WinForms, WPF) · Qt · GTK · Swift · Python · Bash · Kotlin · Jetpack Compose · Rust / Slint · HTML/CSS/JS · Cloudflare · STM32 · ESP32 · Linux · KiCad · Altium · Modbus · CAN · LVGL

## CONTACT

info@viteh.org · viteh.org

This portfolio documents the visible part of the work done since 2014; there were more projects, including internal and tooling work that is not shown. It was first compiled in 2023 from project archives and original photographs, and updated in 2026 with recent work. Historical products, current prototypes and ongoing R&D are identified separately. Images are from the supplied project archive unless otherwise noted.